

VAHAGN GHAZARYAN

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EDUCATION

Shanghai Jiao Tong University | QS #47

Sep 2021 – June 2025

B.Sc. in Software Engineering

Courses: Computer Architecture, Operating Systems, Advanced Data Structures, Cloud Computing, Computer Networking.

WORK EXPERIENCE

IPADS

Systems Engineering Intern

Apr 2024 – June 2025

- Identified performance bottlenecks during co-execution of LLM agents and mobile applications, including up to 237% degradation in UI responsiveness and a 4% increase in energy consumption.
- Developed a pull-based UI state retrieval mechanism using Android Accessibility Service to address slow UI responsiveness.
- Implemented a dynamic background app freezing mechanism using the Linux cgroup freezer subsystem to reduce unnecessary CPU usage.
- Achieved a 15.6% improvement in UI responsiveness and a 3.8% reduction in energy consumption, noticeably improving user experience on mobile devices.

Cognaize

Sep 2019 - Sep 2020

Junior Data Engineer

- Addressed data management complexities caused by different teams using multiple cloud services for storing financial documents.
- Built a unified, user-friendly Python library that wraps and standardizes cloud APIs like AWS S3, Google Cloud Storage, Dropbox, and Google Drive under a single interface.
- Used the library to automate the migration of over 100GB of financial documents from Dropbox to Google Cloud Storage.
- After extensive testing, the library was adopted across the internal codebase, making cloud workflows easier and more consistent.

PROJECTS

ClipOnCloud: Confidential ML Deployment

- Led the deployment of CLIP-based models to AWS EC2 as part of an AI privacy research project.
- Used torch.distributed to coordinate inference across instances.
- Received top evaluation in a technical exhibition, with a paper expected to be published in 2026.
- [GitHub Repository](#)

OHScrcpy: OpenHarmony Screen Mirroring Controller

- Addressed the lack of a high-performance screen mirroring and remote control tool for OpenHarmony devices, enabling remote programming and testing.
- Implemented client-side control using PyQt6, with support for touch, scroll, keyboard input, and dynamic screen resizing; achieved an average of 20 FPS with minimal latency.

- Open-sourced the solution on Gitee and gained recognition from the OpenHarmony community.
- Earned Third Prize at the China Postgraduate Innovation Competition (CPIPC) 2024.
- [Gitee Repository](#)

minik8s: Container Orchestration System

- Designed and developed as a solo coursework project — a lightweight, Kubernetes-inspired system for automating container deployment, scaling, and management.
- Implemented core components including the Control Plane (API Server, Scheduler, Controllers) and Worker Node (Kubelet, KubeProxy).
- Used Docker as the underlying container runtime, Weave for networking, Etcd for state management, and Iptables for traffic control; leveraged GitHub Actions for continuous integration and automated testing.
- [GitHub Repository](#)

Bookstore: Online Shop Platform

- Designed and developed a solo coursework project — a scalable, fault-tolerant e-commerce platform with low-latency performance.
- Implemented advanced features such as role-based authentication, admin dashboards, and sales analytics.
- Used React.js for the frontend and Spring Boot for backend logic; leveraged MySQL, MongoDB, Neo4j, and Redis for different storage needs, and integrated Apache Kafka and Spark for real-time messaging and data processing.
- [GitHub Repository](#)

OPEN SOURCE

- **StringZilla (2.5k+ GitHub stars)** – Enhanced CLI support by adding split/wc functionality.
[GitHub Pull Request](#)

AWARDS

- Third Prize, **China Postgraduate Innovation and Practice Series Competition** Oct 2024
- First Prize, **Hack the DNA Datathon**, organized by ISTC & PMI Science Dec 2019

SKILLS

- **Programming:** Strong scripting and programming skills in Python and Shell; experienced with C/C++, Java and JavaScript.
- **DevOps & Cloud:** Proficiency in Linux and Docker; hands-on experience with AWS (EC2, IAM, S3, CloudWatch); GitHub Actions for CI/CD.
- **Languages:** Strong English communication skills; native in Armenian; conversational in Chinese and Russian.